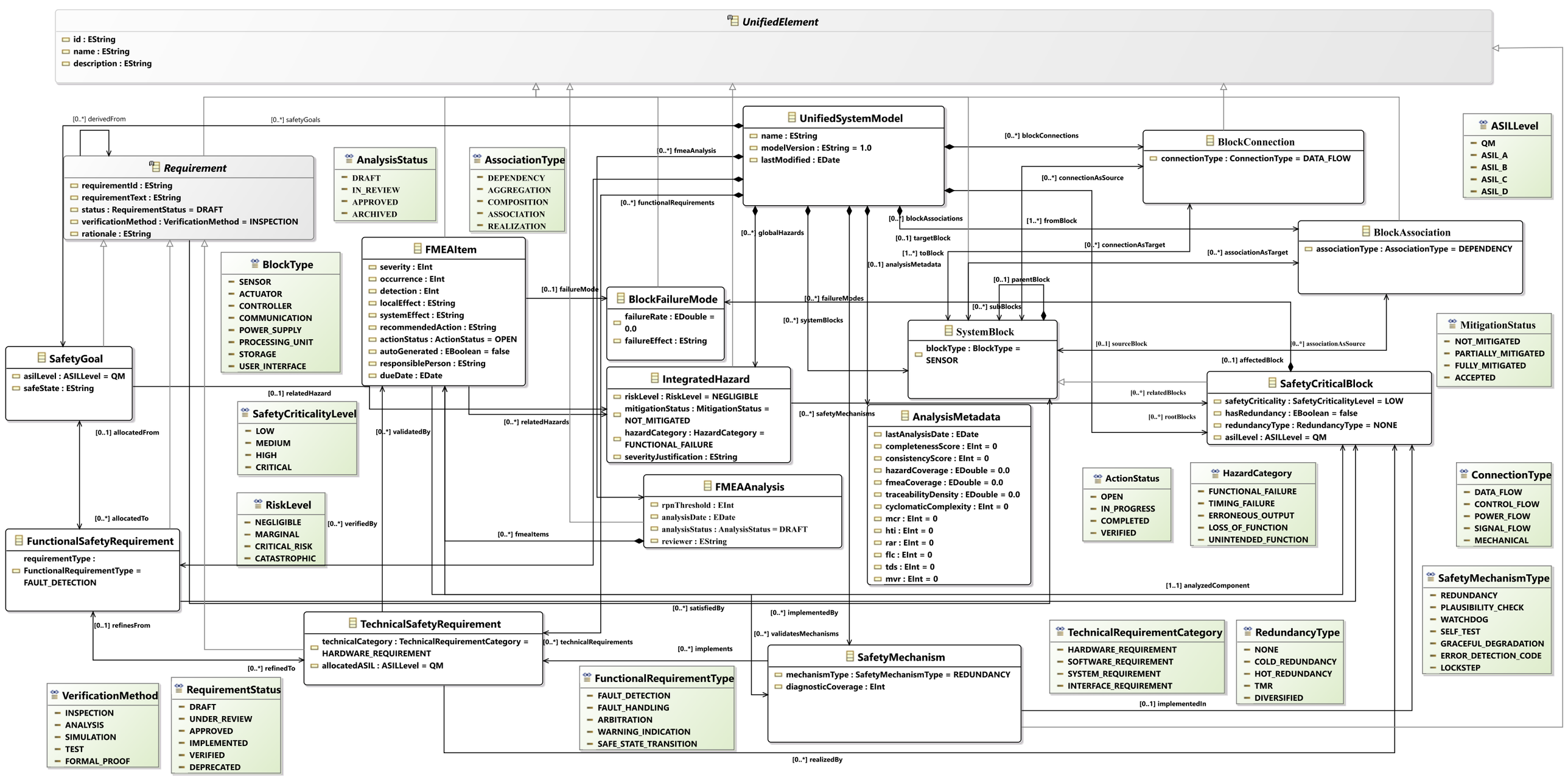
UnifiedElement
id : EString
name : EString
description : EString
Requirement
requirementId : EString
requirementText : EString
status : RequirementStatus = DRAFT
verificationMethod : VerificationMethod = INSPECTION
rationale : EString
UnifiedSystemModel
name : EString
modelVersion : EString = 1.0
lastModified : EDate
BlockConnection
connectionType : ConnectionType = DATA_FLOW
BlockAssociation
associationType : AssociationType = DEPENDENCY
SystemBlock
blockType : BlockType = SENSOR
SafetyCriticalBlock
safetyCriticality : SafetyCriticalityLevel = LOW
hasRedundancy : EBoolean = false
redundancyType : RedundancyType = NONE
asilLevel : ASILLevel = QM
FMEAItem
severity : EInt
occurrence : EInt
detection : EInt
localEffect : EString
systemEffect : EString
recommendedAction : EString
actionStatus : ActionStatus = OPEN
autoGenerated : EBoolean = false
responsiblePerson : EString
dueDate : EDate
BlockFailureMode
failureRate : EDouble = 0.0
failureEffect : EString
IntegratedHazard
riskLevel : RiskLevel = NEGLIGIBLE
mitigationStatus : MitigationStatus = NOT_MITIGATED
hazardCategory : HazardCategory = FUNCTIONAL_FAILURE
severityJustification : EString
FMEAAnalysis
rpnThreshold : EInt
analysisDate : EDate
analysisStatus : AnalysisStatus = DRAFT
reviewer : EString
AnalysisMetadata
lastAnalysisDate : EDate
completenessScore : EInt = 0
consistencyScore : EInt = 0
hazardCoverage : EDouble = 0.0
fmeaCoverage : EDouble = 0.0
traceabilityDensity : EDouble = 0.0
cyclomaticComplexity : EInt = 0
mcr : EInt = 0
hti : EInt = 0
rar : EInt = 0
flc : EInt = 0
tds : EInt = 0
mvr : EInt = 0
SafetyGoal
asilLevel : ASILLevel = QM
safeState : EString
FunctionalSafetyRequirement
requirementType : FunctionalRequirementType = FAULT_DETECTION
TechnicalSafetyRequirement
technicalCategory : TechnicalRequirementCategory = HARDWARE_REQUIREMENT
allocatedASIL : ASILLevel = QM
SafetyMechanism
mechanismType : SafetyMechanismType = REDUNDANCY
diagnosticCoverage : EInt
AnalysisStatus
DRAFT
IN_REVIEW
APPROVED
ARCHIVED
AssociationType
DEPENDENCY
AGGREGATION
COMPOSITION
ASSOCIATION
REALIZATION
BlockType
SENSOR
ACTUATOR
CONTROLLER
COMMUNICATION
POWER_SUPPLY
PROCESSING_UNIT
STORAGE
USER_INTERFACE
SafetyCriticalityLevel
LOW
MEDIUM
HIGH
CRITICAL
RiskLevel
NEGLIGIBLE
MARGINAL
CRITICAL_RISK
CATASTROPHIC
VerificationMethod
INSPECTION
ANALYSIS
SIMULATION
TEST
FORMAL_PROOF
RequirementStatus
DRAFT
UNDER_REVIEW
APPROVED
IMPLEMENTED
VERIFIED
DEPRECATED
FunctionalRequirementType
FAULT_DETECTION
FAULT_HANDLING
ARBITRATION
WARNING_INDICATION
SAFE_STATE_TRANSITION
ASILLevel
QM
ASIL_A
ASIL_B
ASIL_C
ASIL_D
MitigationStatus
NOT_MITIGATED
PARTIALLY_MITIGATED
FULLY_MITIGATED
ACCEPTED
ActionStatus
OPEN
IN_PROGRESS
COMPLETED
VERIFIED
HazardCategory
FUNCTIONAL_FAILURE
TIMING_FAILURE
ERRONEOUS_OUTPUT
LOSS_OF_FUNCTION
UNINTENDED_FUNCTION
ConnectionType
DATA_FLOW
CONTROL_FLOW
POWER_FLOW
SIGNAL_FLOW
MECHANICAL
SafetyMechanismType
REDUNDANCY
PLAUSIBILITY_CHECK
WATCHDOG
SELF_TEST
GRACEFUL_DEGRADATION
ERROR_DETECTION_CODE
LOCKSTEP
TechnicalRequirementCategory
HARDWARE_REQUIREMENT
SOFTWARE_REQUIREMENT
SYSTEM_REQUIREMENT
INTERFACE_REQUIREMENT
RedundancyType
NONE
COLD_REDUNDANCY
HOT_REDUNDANCY
TMR
DIVERSIFIED
[0..*] derivedFrom
[0..*] safetyGoals
[0..*] fmeaAnalysis
[0..*] functionalRequirements
[0..*] blockConnections
[0..*] connectionAsSource
[0..*] blockAssociations
[1..*] fromBlock
[0..1] targetBlock
[1..*] toBlock
[0..1] analysisMetadata
[0..*] connectionAsTarget
[0..*] associationAsTarget
[0..*] globalHazards
[0..1] parentBlock
[0..*] subBlocks
[0..1] failureMode
[0..*] failureModes
[0..*] systemBlocks
[0..1] sourceBlock
[0..*] associationAsSource
[0..1] affectedBlock
[0..1] relatedHazard
[0..*] relatedBlocks
[0..*] rootBlocks
[0..*] relatedHazards
[0..*] safetyMechanisms
[0..1] allocatedFrom
[0..*] validatedBy
[0..*] verifiedBy
[0..*] allocatedTo
[0..*] fmeaItems
[1..1] analyzedComponent
[0..*] satisfiedBy
[0..*] implementedBy
[0..1] refinesFrom
[0..*] validatesMechanisms
[0..*] technicalRequirements
[0..*] implements
[0..*] refinedTo
[0..1] implementedIn
[0..*] realizedBy